

Causal version of Shapley values

Overview

Shapley values as a sum over permutations π

$$Sh_j = \sum_{\pi} w(\pi) [\nu(S \cup \{j\}) - \nu(S)]$$

where

$$\phi_j(\pi) := \nu(\{\ell : \ell \preceq_\pi j\}) - \nu(\{\ell : j \prec_\pi \ell\}),$$

$w(\pi)$ is a weight for each permutation, and ν is a given value function

Overview

Variant	$\nu(S)$	Weakness
Conditional	$\mathbb{E}[f(X) \mid X_S = x_S]$	Association, not causation
Marginal	$\mathbb{E}[f(x_S, X_{\bar{S}})]$	Ignores links among features
Causal (Heskes)	$\mathbb{E}[f(X) \mid \text{do}(X_S = x_S)]$	Needs a causal graph
do-Shapley (Jung)	$\mathbb{E}[Y \mid \text{do}(X_S = x_S)]$	Needs a graph, targets Y

Every variant is the same formula. Only $\nu(S)$, how the features in S are fixed, changes.

Causal Shapley values (Heskes, 2020)

Definition

- ▶ Explains one prediction of a model f .
- ▶ Features are fixed with $\text{do}(\cdot)$ using the causal graph
- ▶ Uniform weight over all permutation in the formula leads to a symmetric version of Shapley values
- ▶ If we impose to respect the causal structure we have an asymmetric

Causal Shapley values (Heskes, 2020)

Decomposing Shapley values into direct and indirect effects

For a fixed permutation π , $\phi_j(\pi)$ reads as

$$\mathbb{E}[f(X_{\bar{S}}, x_{\underline{S} \cup j}) | do(X_{\underline{S} \cup j} = x_{\underline{S} \cup j})] - \mathbb{E}[f(X_{\bar{S} \cup j}, x_{\underline{S}}) | do(X_{\underline{S}} = x_{\underline{S}})]$$

It can be written as the sum of a direct effect

$$\mathbb{E}[f(X_{\bar{S}}, x_{\underline{S} \cup j}) | do(X_{\underline{S}} = x_{\underline{S}})] - \mathbb{E}[f(X_{\bar{S} \cup j}, x_{\underline{S}}) | do(X_{\underline{S}} = x_{\underline{S}})]$$

and an indirect effect

$$\mathbb{E}[f(X_{\bar{S}}, x_{\underline{S} \cup j}) | do(X_{\underline{S} \cup j} = x_{\underline{S} \cup j})] - \mathbb{E}[f(X_{\bar{S}}, x_{\underline{S} \cup j}) | do(X_{\underline{S}} = x_{\underline{S}})]$$

where $\underline{S} = \{\ell, \ell \prec j\}$ and $\bar{S} = \{\ell, j \succ \ell\}$

Causal Shapley values (Heskes, 2020)

Decomposing Shapley values into direct and indirect effects

- ▶ direct effect: expected change in prediction when X_j replaced x_j without changing the distribution of the other out-of-coalition features
- ▶ indirect effect : difference in expectation when the distribution of the other out-of-coalition features changes due to the additional intervention $do(X_j = x_j)$
- ▶ direct and indirect parts of Shapley values computed as average over all permutations

Causal Shapley values (Heskes, 2020)

Decomposing Shapley values into direct and indirect effects

- ▶ Conditional Shapley values can be decomposed similarly by replacing conditioning by intervention with conditioning by observation.
- ▶ For marginal Shapley values, there is no conditioning and hence no indirect effect: by construction marginal Shapley values can only represent direct effects

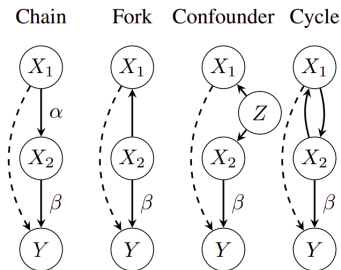
Causal Shapley values (Heskes, 2020)

Some toy examples

- ▶ Assume that we have two features X_1, X_2 with $\mathbb{E}[X_2|X_1] = \alpha X_1$ and $E[X_2] = \mathbb{E}[X_2] = 0$
- ▶ Set $f(x_1, x_2) = \beta x_2$
- ▶ It may correspond to several causal structures : chain, fork, confounder or cycle

Causal Shapley values (Heskes, 2020)

Some toy examples



An example from Heskes (2020)

See `CausalExamples.ipynb`

Definition of do-Shapley effects (Jung et al. 2022)

Same value function

- ▶ As causal Shapley
- ▶ Applied to the real outcome Y instead of a model f :
$$\nu(S) = [Y \mid \text{do}(X_S = x_S)].$$

Definition of do-Shapley effects (Jung et al. 2022)

Contributions of Jung et al.

- ▶ Axiomatize causal contribution measures
- ▶ Conditions under which the do Shapley values can be inferred from the observational data (identifiability) in polynomial time (computational feasibility)
- ▶ Construct estimators for the do-Shapley, exhibiting consistency, computational feasibility, and statistical robustness

Definition of do-Shapley effects (Jung et al. 2022)

Setting

- ▶ SCM : $\mathcal{M}$
- ▶ Sample drawn from a distribution $\mathbb{P}_M$
- ▶ Distribution compatible with a graph $\mathcal{G}$ induced by the SCM on topologically ordered variables $(\mathbf{V}, Y)$, with Y supposed to be the final variable in the order.

Definition of do-Shapley effects (Jung et al. 2022)

Goals

- ▶ Define the concept of contribution of $v_i \in \mathbf{V}$ to the target causal effect

$$\mathbb{E}[Y | do(\mathbf{V} = \mathbf{v})]$$

- ▶ If Y is generated by a deterministic function, $Y := f(V)$, then the task reduces to define a concept of causal contribution of the features v_i to the ML model
- ▶ Estimation of these quantities?

Definition of do-Shapley effects (Jung et al. 2022)

The definition of causal contributions has to satisfy some axioms

Axioms for Causal Contribution

Causal contributions $\phi_{v_1}, \dots, \phi_{v_k}$ of each variable $v_1, \dots, v_k$ w.r.t. G is considered desirable if the following properties are satisfied:

- **Perfect assignment.** Contributions are perfectly assigned

$$\mathbb{E}[Y|do(\mathbf{V} = \mathbf{v})] = \sum_{i=1}^k \phi_{v_i}$$

- **Causal irrelevance.** If V_i is causally irrelevant to Y for all witness $\mathbf{w}$, that is $\mathbb{P}[Y|do(V_i = v_i, \mathbf{W} = \mathbf{w})] = \mathbb{P}[Y|\mathbf{W} = \mathbf{w}]$, then $\phi_{v_i} = 0$.

Definition of do-Shapley effects (Jung et al. 2022)

Axioms for Causal Contribution

- **Causal symmetry.** If (V_i, V_j) have the same causal explanatory power to Y , that is if for all $\mathbf{W}$,

$$\mathbb{P}[Y|do(V_i, \mathbf{W}) = \mathbf{w}] = \mathbb{P}[Y|do(V_j, \mathbf{W}) = \mathbf{w}]$$

then $\phi_{v_i} = \phi_{v_j}$

- **Causal approximation.**

Definition of do-Shapley effects (Jung et al. 2022)

Comments

- ▶ Perfect assignment is a natural requirement since the aim is to attribute the degree of contributions of each feature v_i to the target causal effect.
- ▶ Causal irrelevance reflects a desire to understand the cause of the outcome by forcing zero contributions for variables not causing the outcome.
- ▶ Causal symmetry enforces the equal contribution for a pair of features if they have the same causal explanatory power.
- ▶ Causal approximation allows ϕ_{v_i} to be interpreted as a proxy for the causal effect s.t

Definition of do-Shapley effects (Jung et al. 2022)

Definition

The do-Shapley value is

$$\phi_{v_i} = \sum_S \omega(S) [\mathbb{E}[Y | do(\mathbf{V}_{S,i} = \mathbf{v}_{S,i})] - \mathbb{E}[Y | do(\mathbf{V}_S = \mathbf{v}_S)]]$$

where $\omega(S) = 1/n \times \binom{n-1}{|S|}^{-1}$

Definition of do-Shapley effects (Jung et al. 2022)

Theorem (Uniqueness of the do-Shapley)

The doShapley is a unique causal contribution measure satisfying all the properties state in the Axioms for causal contributions